QUESTION BANK

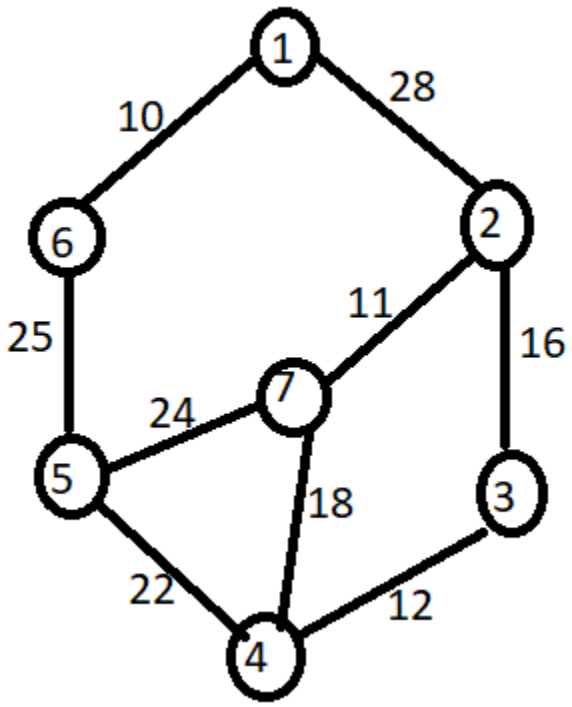
Course: ADVANCED DATA STRUCTURES AND ALGORITHM ANALYSIS

Branch: IT & CS&BS

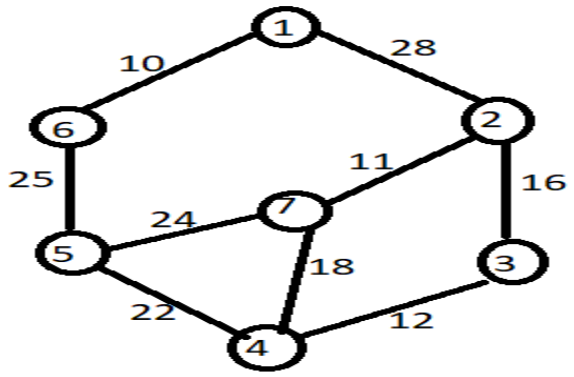
UNIT-III

1. Write an algorithm for job sequencing with deadlines using greedy method and find an optimal job schedule for n=5, profits (P1, P2, P3, P4, P5)=(60, 100, 20 40, 65) and deadlines (d1, d2, d3, d4, d5) =(2, 1, 3, 2, 1)

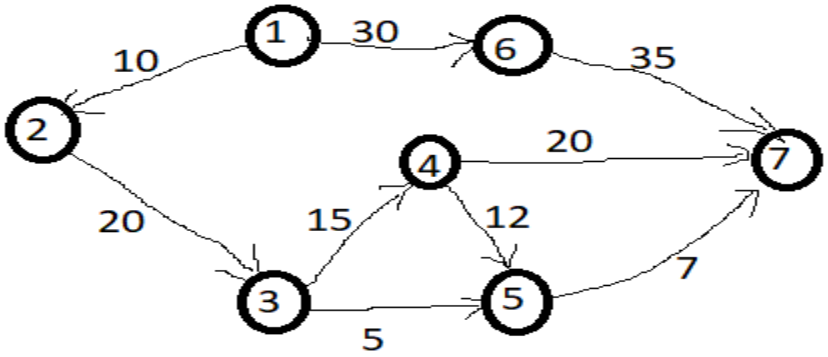
2. Find the optimal solution for knapsack problem using greedy method for n=5, m=60, profits (p1, p2, p3, p4, p5) = (30, 40, 45, 77, 90) and weights (w1, w2, w3, w4, w5) = (5, 10, 15, 22, 25)
3. Find minimum cost spanning tree for the following graph using kruskal's algorithm.



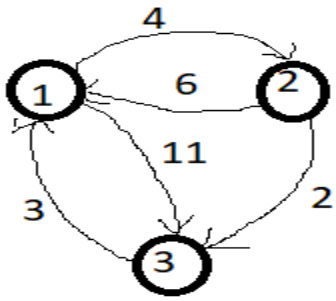
4. Find the minimum cost spanning tree for the following graph using prim's algorithm.



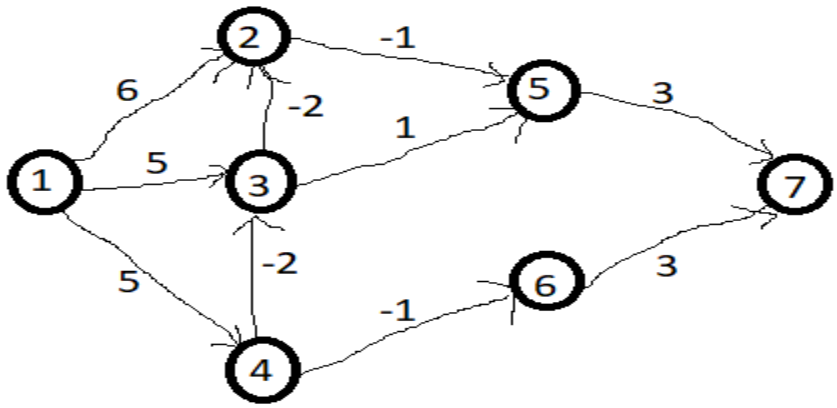
5. Write Prim's algorithm to find minimum cost spanning tree.
6. Find shortest paths to all the vertices from source vertex in the following graph using Dijkstra algorithm.



7. Find the shortest paths between all pairs of vertices in the following graph.



8. Find the shortest paths from source vertex to all the remaining vertices in the following graph using dynamic programming.



9. Construct OBST for n=4, (a1, a2, a3, a4)= (do, if, int, while), p(1:4)=(3, 3, 1, 1) and q(0:4)=(2, 3, 1,1 ,1)

10. Solve 0/1 knapsack problem using dynamic programming for knapsack instance n=3, m=6, (p1,p2,p3)=(1,2,5) and (w1,w2,w3)=(2,3,4).

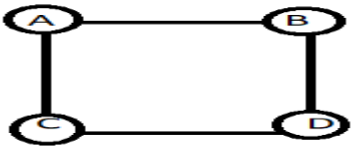
11. Find the shortest route for travelling sales person problem by using the following cost matrix.

$$\begin{bmatrix} 0 & 10 & 15 & 20 \\ 5 & 0 & 9 & 10 \\ 6 & 13 & 0 & 12 \\ 8 & 8 & 9 & 0 \end{bmatrix}$$

12. Write an algorithm to construct OBST.

UNIT-IV

1. Solve 4-queens problem using back tracking.
2. Write an algorithm to solve N-Queens problem using back tracking.
3. Solve sum of subsets problem for n=4, m=31 and w[1:4]=(7,11,13,24)
4. solve the graph coloring problem for the following graph and m=3.



5. Write an algorithm for graph coloring.
6. Solve 0/1 knapsack problem using FIFO branch and bound for knapsack instance n=4,m=15, $(p_1,p_2,p_3,p_4)=(10,10,12,18)$ and $(w_1,w_2,w_3,w_4)=(2,4,6,9)$.
7. Construct state space tree for 0/1 knapsack problem using Least cost branch and bound for knapsack instance n=4,m=15, $(p_1,p_2,p_3,p_4)=(10,10,12,18)$ and $(w_1,w_2,w_3,w_4)=(2,4,6,9)$.
8. Find optimal tour for the following cost matrix using Least Cost branch and bound.

$$\begin{bmatrix} \infty & 20 & 30 & 10 & 11 \\ 15 & \infty & 16 & 4 & 2 \\ 3 & 5 & \infty & 2 & 4 \\ 19 & 6 & 18 & \infty & 3 \\ 16 & 4 & 7 & 16 & \infty \end{bmatrix}$$

9. Find optimal tour for the following cost matrix using Least cost branch and bound.

$$\begin{bmatrix} \infty & 5 & 4 & 7 \\ 6 & \infty & 2 & 1 \\ 9 & 7 & \infty & 9 \\ 5 & 9 & 8 & \infty \end{bmatrix}$$

UNIT-V

1. Write about P, NP, NP-hard and NP-complete problems and their relationship.
2. Explain Cook's theorem.
3. Explain clique decision problem.
4. Explain chromatic number decision problem and prove that CNDP is NP-hard.
5. Explain travelling sales person decision problem.
6. Explain scheduling identical processors with an example.